

■ Python + Systems Logic Roadmap

This roadmap blends Python fundamentals with systems-level reasoning. It's designed to build engineering intuition—understanding not only *how* code runs, but *why* systems behave the way they do.

1■■ Foundation — Core Python Mechanics

- Understand data types and variables: strings, lists, dictionaries, tuples.
- Learn flow control: if/else statements, for and while loops.
- Grasp functions and scope: parameters, return values, and local/global logic.
- Use try/except for error handling (like ImportError handling in `test_lightglue_import.py`).
- Explore Python's object model and basic file handling.

2■■ Application — Systems Integration Layer

- Practice automation scripts: moving files, reading configs, using environment variables.
- Understand how virtual environments isolate runtime dependencies.
- Integrate libraries like NumPy, OpenCV, and Matplotlib to interact with real data.
- Build small diagnostic utilities to inspect your system (e.g., listing installed packages or verifying paths).
- Simulate process flow: use logging and command-line arguments to track script execution.

3■■ Integration — Vision & Neural Systems

- Combine PyTorch and LightGlue to form a minimal vision pipeline.
- Use SuperPoint and LightGlue to detect, match, and visualize keypoints between images.
- Learn model inference flow: from tensors → features → matches → visualization.
- Track dependencies through imports and environment structure.
- Document findings in Markdown and commit results with Git.

■ Terminal Session Logs

Use this section to paste key terminal outputs or environment verifications, such as successful installations, test script results, or troubleshooting logs. Example: running ``python test_lightglue_import.py`` or showing pip freeze results.

Example entry:

■ All imports successful. LightGlue and dependencies are ready.

"Understanding systems is the first step toward mastering them."